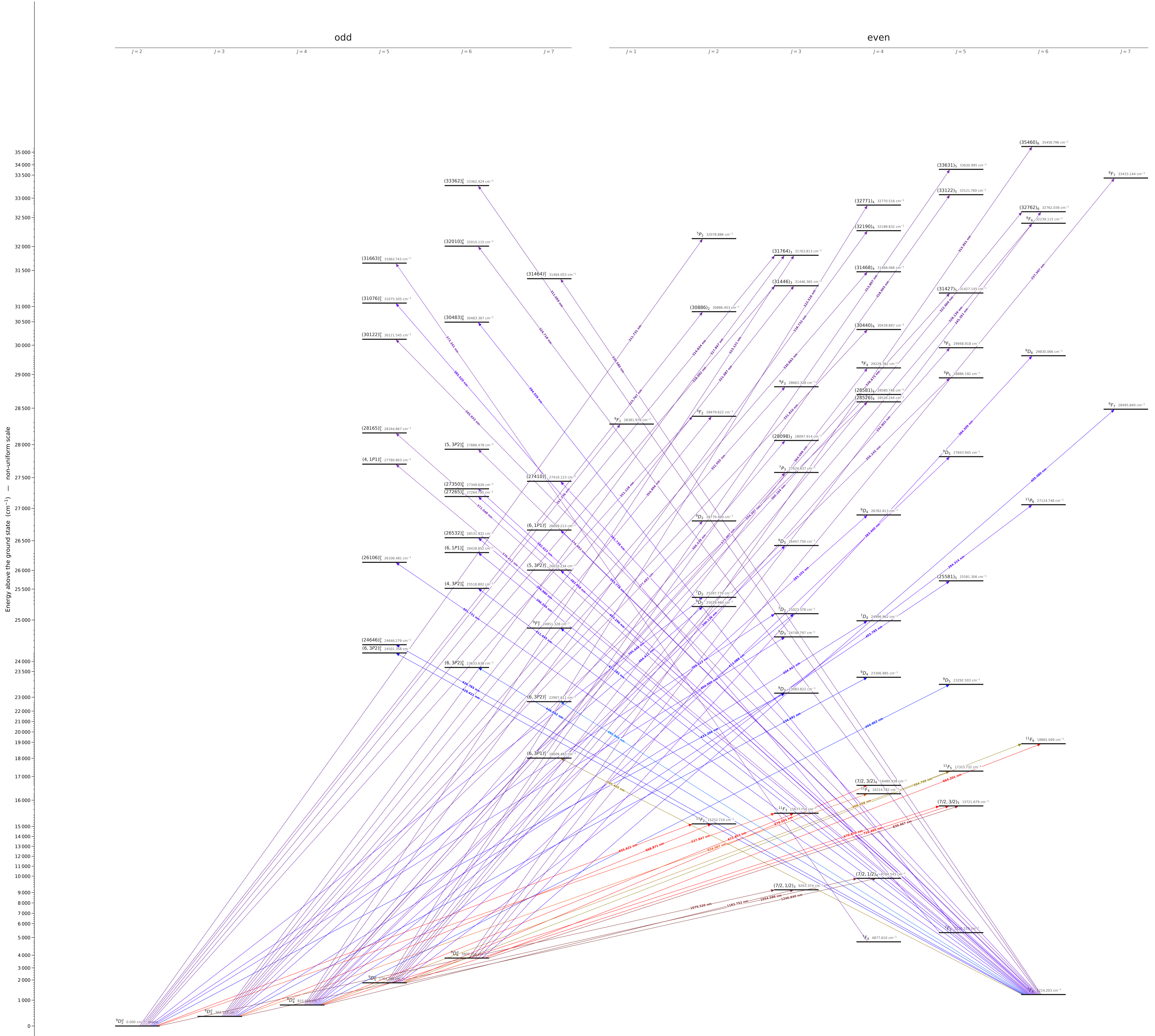
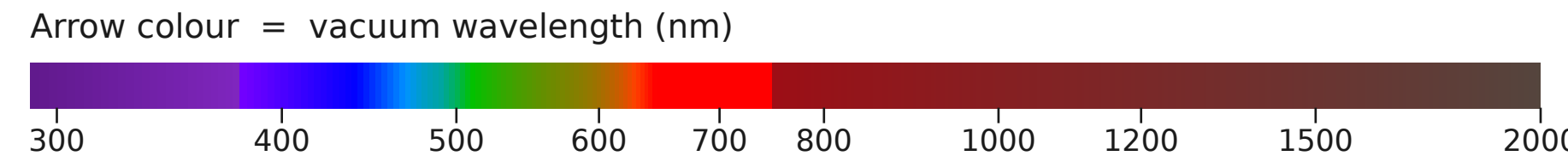


Curium

Cm (Cm I). Classified transitions below 2 μm (levels and lines from the literature): 83 levels, 88 lines.

Cm⁺ ionisation limit: 48330.68 cm⁻¹ ≈ 5.99224 eV (206.908 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U|J\pi|U'|$ in ea₀
Matrix elements are measured or NIST values: * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U J\pi U' $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
⁹ D ₂ - ¹ D ₂	395.4481	395.3362	758.1083	-	-	-	-	first step of Mainz scheme C
⁹ D ₁ - (²⁵⁵⁸¹) ₅	403.7851	403.6710	742.4555	-	-	-	-	excitation from the thermally populated 9D4 level seen in...
⁹ D ₂ - ⁹ D ₃	404.0600	403.9459	741.9503	-	-	-	-	first step of Mainz scheme B
⁹ D ₂ - ⁹ D ₂	404.4112	404.2970	741.3060	-	-	-	-	excitation from the thermally populated 9D3 level seen in...
⁹ D ₂ - ¹ D ₁	404.9434	404.8290	740.3318	-	-	-	-	excitation from the thermally populated 9D3 level seen in...
⁹ D ₂ - ⁹ D ₃	433.2038	433.0820	692.0356	-	-	-	-	first step of Mainz two-step resonance-ionization scheme...

λ , ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, 1/(2 $\pi\tau$).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁹ F ₆ - (³³³⁶²) ₆	311.059	246-244	6446.00
⁹ D ₂ - ⁹ D ₂	311.731	246-244	-14210.00
⁹ D ₂ - (³³⁶³¹) ₅	313.807	246-244	-1199.00
⁹ D ₂ - (³¹⁷⁶⁴) ₃	314.824	246-244	-10073.00
⁹ D ₂ - (³⁵⁴⁶⁰) ₆	315.951	246-244	-11812.00
⁹ D ₂ - (³¹⁷⁶⁴) ₃	317.847	246-244	-10103.00
⁹ D ₂ - (³¹⁴⁴⁶) ₃	318.002	246-244	-10403.00
⁹ D ₂ - (³²¹⁹⁰) ₄	318.733	246-244	-6595.00
⁹ D ₂ - (³³¹²²) ₅	318.903	246-244	-7675.00
⁹ D ₂ - (³¹⁴⁶⁸) ₄	320.863	246-244	-11392.00
⁹ D ₂ - (³¹⁴⁴⁶) ₃	321.087	246-244	-10433.00
⁹ D ₂ - (³²⁷⁷¹) ₄	322.516	246-244	-12052.00
⁹ D ₂ - (³²⁷⁶²) ₆	322.604	246-244	-10613.00
⁹ D ₁ - (³¹⁷⁶⁴) ₃	323.121	246-244	-10133.00
⁹ D ₂ - (³⁰⁸⁸⁶) ₂	323.767	246-244	-13431.00
⁹ F ₆ - (³³⁰¹⁰) ₆	324.718	246-244	4317.00
⁹ D ₂ - (³¹⁴²⁷) ₅	326.675	246-244	-10972.00
⁹ D ₂ - ⁹ F ₆	328.139	246-244	-10193.00
⁹ F ₆ - (³¹⁴⁶⁴) ₅	330.580	246-244	4647.00
⁹ D ₂ - (³⁰⁴⁴⁰) ₄	331.810	246-244	-12771.00
⁹ D ₂ - ⁹ F ₇	337.567	246-244	-11272.00
⁹ D ₂ - (³²⁷⁶²) ₆	345.391	246-244	-10852.00

Shift = ν (first isotope) - ν (second isotope).